

ABOUT US

Octavus is a global business intelligence consulting, specialized in providing holistic market solutions for diverse Healthcare and Life sciences sector including Pharma, Biopharma, MedTech, Nutraceuticals, Cosmetics, Diagnostics, Digital health, Animal health, HealthTech and Healthcare start-ups

INSIGHTS CORNER

WHAT IS INSIGHTS CORNER

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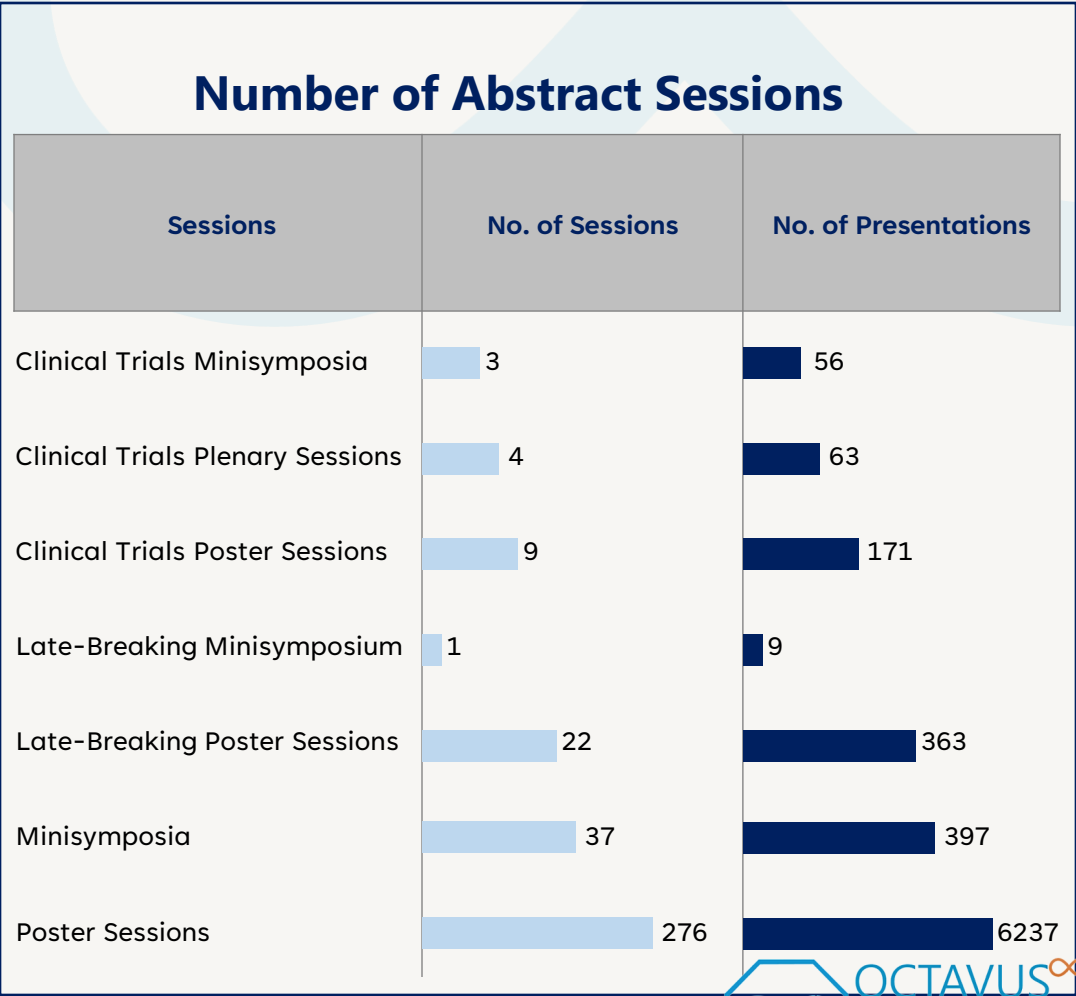


AGENDA

Objectives:

- Highlight the most impactful scientific advances, novel therapies, and emerging technologies presented at AACR 2025
- Summarize key trends in cancer research, clinical trials, and drug development, including the role of artificial intelligence
- Share actionable insights and future outlooks for oncology professionals and stakeholders

Key Sections	
•	Introduction & AACR 2025 Overview
•	Major Scientific Highlights & Breakthroughs
•	New Drugs, Clinical Trials & Technologies
•	Artificial Intelligence in Cancer Care
•	Trends, Future Outlook & Q&A



AACR 2025 ANNUAL MEETING (25-30 APR)

*"Unifying
Cancer Science
and Medicine: A
Continuum of
Innovation for
Impact"*

Proffered Abstracts

7000+

Exhibitors

600+

Educational Sessions

65

Major Symposia

50

Plenary Sessions

6

ABSTRACTS CATEGORY

1

Bioinformatics, Computational Biology,
Systems Biology, and Convergent Science

2

Chemistry

3

Clinical Research: Solid Tumors and
Hematologic Malignancies

4

Clinical Trials: Solid Tumors and
Hematologic Malignancies

5

Endocrinology

6

Experimental and Molecular Therapeutics

7

Immunology

8

Molecular/Cellular Biology and Genetics

9

Population Sciences

10

Prevention, Early Detection, and
Interception

11

Regulatory Science and Policy

12

Science and Health Policy

13

Tumor Biology

DAY-1 26TH-APR

1

First major scientific session, "**Discovery Science Plenary:** Novel Mechanisms Influencing Cancer Evolution"

2

Over 65 Educational Sessions and Methods Workshops concluded

3

The AACR Runners for Research 5K Run/Walk saw over 850 participants and raised more than \$273,000 for cancer research

4

Two key sessions addressed

DAY-2 27TH-APR

1

The AACR announced its largest-ever grant initiative

2

Opening Plenary: Unifying Cancer Science and Medicine

3

Clinical Trials Program: Immunotherapy and Targeted Therapy Advances

- KEYNOTE-689 Study: Pembrolizumab before and after surgery in HNSCC
- dMMR Tumors: PD-1
- Combination Therapy: PD-1 immunotherapy with the PARP inhibitor (Olaparib)
- Early data on a Werner helicase inhibitor for MSI/dMMR tumors

4

Two sessions unveiled eight novel therapeutic molecules, including:

- New RAS pathway inhibitors (AMG 410, RMC-5127)
- Immune-targeting agents (HER2 x CD3 engager, FAP-targeting drugs)
- Androgen receptor and transcription factor inhibitors, among others

5

Artificial Intelligence and Machine Learning in Cancer Research

6

Sessions explored the lifestyles impact on cancer risk

DAY-3 28TH-APR

1

Plenary Session: Innovative Technologies Driving Cancer Research

- Organoid Models
- CAR T Cell Engineering
- Liquid Biopsies
- Decentralized Clinical Trials

2

Clinical Trials Plenary: New Therapies for Lung Cancer

- HER2 Inhibitor Zongertinib
- EGFR Inhibitor Aumolertinib
- ROS1 Inhibitor JYP0322
- TIGIT Antibody Tiragolumab

3

New Drugs on the Horizon

- BAY 3547926
- M0324
- AZD0022
- GSK4418959 (IDE275)

4

Advances in Liquid Biopsy

- The VICTORI trial

5

Hematologic Malignancies: AML Atlas

6

Rare Tumors and AI

DAY-4 29TH-APR

1

Plenary Session: Targeting the Cancer Ecosystem

- Tumor Heterogeneity & Immunogenicity
- Immune System Archetypes
- Spatial Heterogeneity
- Metastatic Evolution

2

Special Session: Overcoming Hurdles in Therapeutic Translation

- Using predictive biomarkers to select patients most likely to benefit or develop resistance.
- Improving tumor sampling technologies to better capture heterogeneity.
- Employing digital twins and virtual trials to model drug effects.
- Emphasizing the value of learning from negative results and optimizing investment decisions in experimental drugs

3

Clinical Trials Plenary: Biologics and T-cell Engagers

- CD40 Agonist Antibody
- HER2/CD3 Bispecific Antibody (Runimotamab)
- Invikafusp Alfa
- ICAM-1 CAR T-cell Therapy (AIC100)

4

Advances in Cancer Vaccines

- New approaches for identifying and targeting neoantigens
- Personalized mRNA neoantigen vaccines
- Novel delivery methods

5

Poster Sessions: Breast Cancer Mortality and Survivorship

6

Combined Liquid and Tissue Biopsy for Tailored Therapy

- The phase II ROME trial demonstrated using both tissue and liquid biopsies

7

Developmental Origins of Childhood Cancer

NEW ONCOLOGY DRUGS REVEALED

COMPANY	DRUGS	MoA	Indication
	AMG 410	pan-KRAS inhibitor	CRC, PDAC, NSCLC
 A Member of the Roche Group	GDC-2992	AR degrader	Prostate cancer
	ABBV-969	Bispecific PSMA/STEAP1 antibody drug conjugate	Prostate cancer
	ABP-102/CT-P72	HER2 x CD3 bispecific T cell engager	Breast cancer, Gastric cancer
	BMS-986449	CELMoD molecular glue degrader of IKZF2/4	Solid Tumors
	RMC-5127	KRAS G12V RAS(ON) tri-complex inhibitor	PDAC and NSCLC
	FXX489	FAP-targeting 177Lu radioligand	PDAC, NSCLC, Breast and Colorectal cancer
	RO7567132	FAP x LTBR bispecific antibody	Solid Tumors
	BAY 3547926	GCP3-targeted 225Ac-radioligand antibody	Hepatocellular carcinoma
	GSK4418959 (IDE275)	WRN helicase inhibitor	CRC and Endometrial cancer
	AZD0022	KRAS G12D inhibitor	CRC, PDAC and NSCLC
	M0324	MUC1/CD40 bispecific antibody	Solid Tumors

2025 NEXTGEN STARS

Researchers recognized for their innovative cancer research	Universities	Cancer Research Topics
Edmond Chan, MD	Columbia University	Exploring dysregulated ribosomal homeostasis in 9p21.3 deleted cancers and microsatellite unstable cancers
Yash Chhabra, PhD	Fox Chase Cancer Center	Investigating age- and sex-differential changes in the tumor microenvironment to inhibit metastasis and enhance therapy responses in melanoma
Karen O. Dixon, PhD	University of Basel	Studying PI5P4K as a molecular checkpoint of anti-tumor immunity
Adam D. Durbin, MD, PhD	St. Jude Children's Research Hospital	Developing a genome-derived non-coding reporter to dissect and control intratumoral heterogeneity and chemoresistance
Anniina Farkkila, MD, PhD	University of Helsinki	Mapping chemotherapy-induced myeloid-driven spatial exhaustion in ovarian cancer and targeting the NECTIN2-TIGIT interaction
Anand D. Jeyasekharan, MBBS, PhD	National University of Singapore	Investigating lymphoma cell spatial organization with MYC-BCL2 co-expression and its impact on immune microenvironment
Zuzana Keckesova, PhD	Institute of Organic Chemistry and Biochemistry	Exploring cancer-resistant tissues for novel tumor suppressor identification
John R. Prensner, MD, PhD	University of Michigan	Mapping the functional microproteome in cancer
Elvin Wagenblast, PhD	Icahn School of Medicine at Mount Sinai	Studying developmental dependency and metabolic vulnerabilities in NUP98-rearranged pediatric leukemia
Stephanie Z. Xie, PhD	University Health Network	Investigating inflammatory memory and selective advantage in human clonal hematopoiesis
Di Zhao, PhD	MD Anderson Cancer Center	Decoding actionable tumor suppressor defects for personalized cancer medicine

ABP-102/CT-P72: PRECLINICAL DATA



ABP-102/CT-P72, a tetravalent bispecific HER2 x CD3 T-cell engager

ABP-102/CT-P72 is product from strategic partnership of Abpro and Celltrion for worldwide development and commercialization

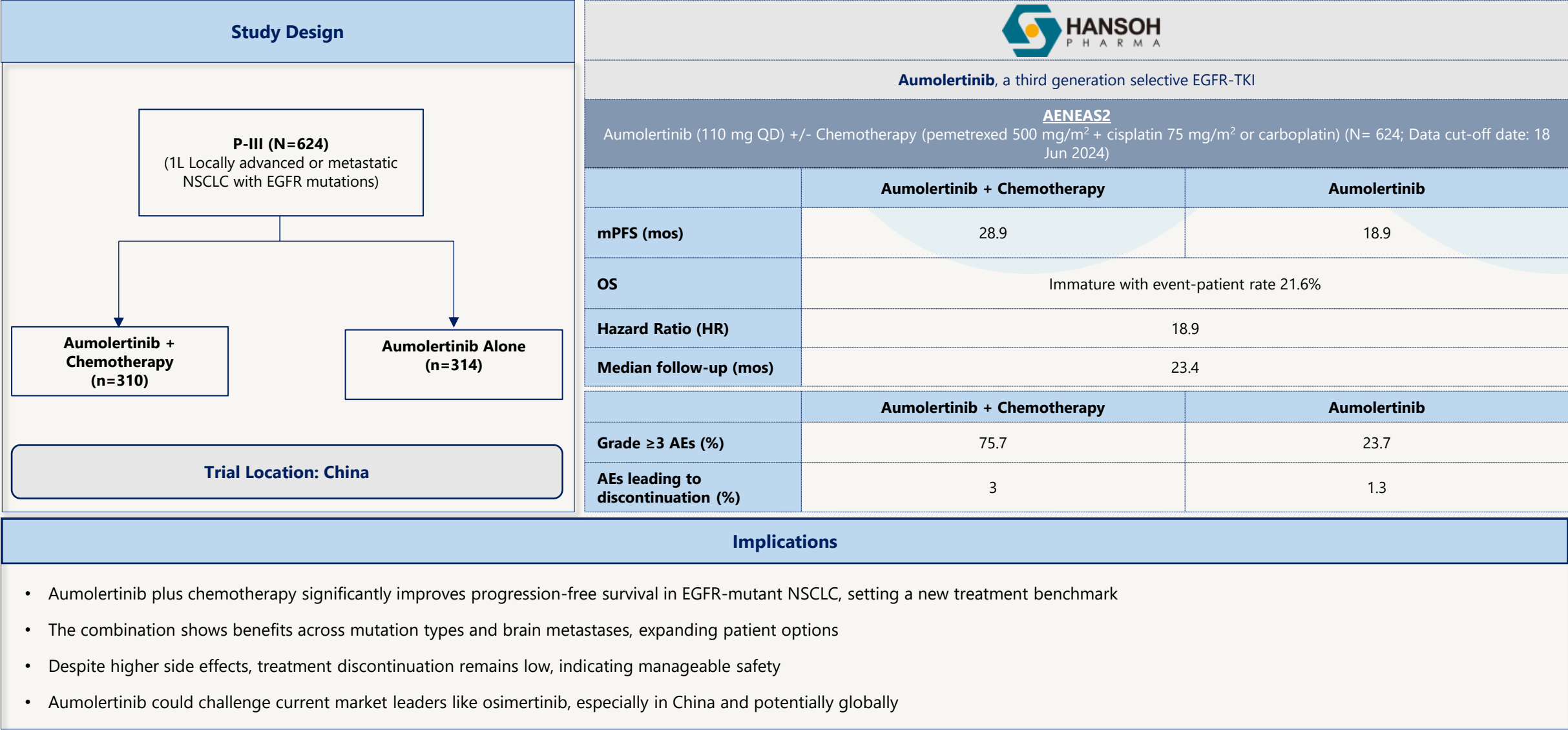
Key Findings	
Highly Selective Tumor Killing	ABP-102/CT-P72 effectively targets HER2-overexpressing breast and gastric cancers while minimizing impact on HER2-low cells, overcoming a major drawback of previous HER2-targeted T-cell engagers
Enhanced Tumor Growth Inhibition	In vivo studies demonstrated ABP-102/CT-P72 achieved up to twice the tumor suppression of a runimotamab biosimilar, a leading HER2xCD3 bispecific antibody
Reduced Cytokine Release	ABP-102/CT-P72's monovalent CD3 binding reduces cytokine toxicity by lowering cytokine release in HER2-low environments while preserving strong tumor-killing effects in HER2-high cancers
Improved Tolerability	Dose escalation studies in cynomolgus monkeys showed ABP-102/CT-P72 was well tolerated at 180 times the maximum tolerated dose of the parental antibody, indicating a broader therapeutic window

Implications
- ABP-102/CT-P72's strong safety and efficacy profile could make it a leading HER2-targeted therapy, expanding treatment options for HER2-positive cancers - The broader therapeutic window may allow higher dosing and better patient tolerance, increasing its market potential; trial initiation is expected in 1H'26 - Partnership with Celltrion supports faster development and global commercialization efforts - Positive preclinical data may boost investor confidence and put competitive pressure on existing HER2 therapies

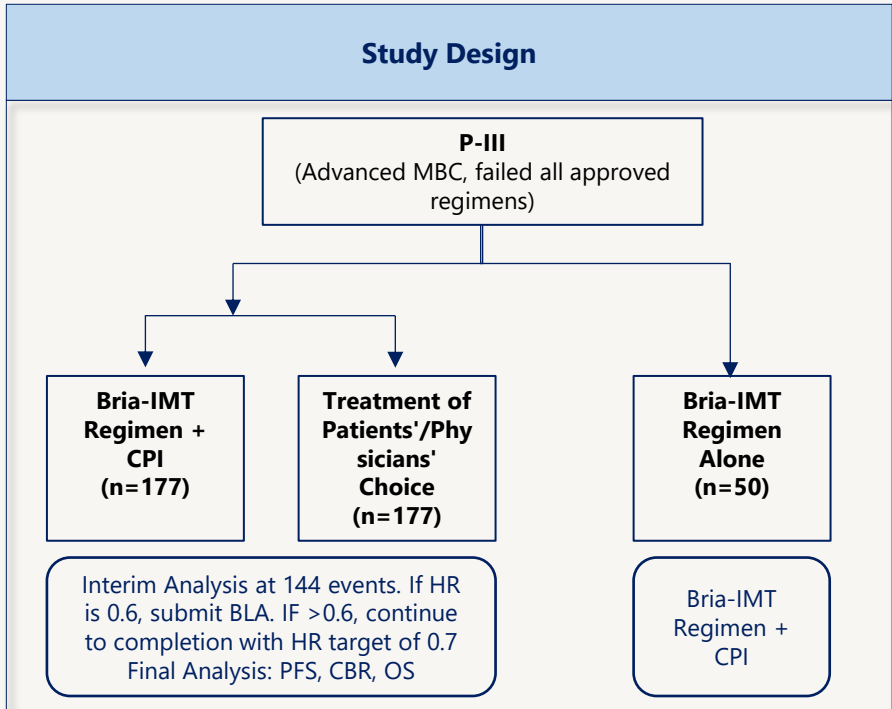
AIC100: P-I DOSE-ESCALATION STUDY RESULTS

Study Design																																			
P-I (N=24) (Newly diagnosed and R/R anaplastic thyroid cancer (ATC) and R/R poorly differentiated thyroid cancer (PDTC)) AIC100 at 1x 10⁷ AIC100 at 1x 10⁸ AIC100 at 5x 10⁸ AIC100 at 7.5x 10⁸ 24 patients enrolled of which 15 (8 ATC; 7 PDTC) were infused with AIC100 at 4DLs 1 Endpoint: Safety assessment/DLTs within 30 days of AIC100 infusion and recommended P-II dose (RP2D)		AIC100, a third generation ICAM-1 directed CAR T-cell product engineered with affinity-tuned technology to selectively bind and kill tumor cells to improve safety NCT04420754 AIC100 (at dose levels (DL) 1x 10⁷, 1 x 10⁸, and 5 x 10⁸) (n= 15; Data cut-off date: 12 Dec 2024) <table><tr><th></th><th>DL1: AIC100 at 1x 10⁷</th><th>DL2: AIC100 at 1x 10⁸</th><th>DL3: AIC100 at 5x 10⁸</th><th>DL4: AIC100 at 7.5x 10⁸</th></tr><tr><td>DLTs</td><td colspan="3">0</td><td>-</td></tr><tr><td>ORR at day 42 after infusion (%)</td><td>0</td><td colspan="2">22</td><td>-</td></tr><tr><td>ORR for ATC pts (%)</td><td>0</td><td colspan="2">50</td><td>-</td></tr><tr><td>DCR at day 42 after infusion (%)</td><td>0</td><td colspan="2">56</td><td>-</td></tr><tr><td>DCR for PDTC pts (%)</td><td>0</td><td colspan="2">60</td><td>-</td></tr></table>					DL1: AIC100 at 1x 10 ⁷	DL2: AIC100 at 1x 10 ⁸	DL3: AIC100 at 5x 10 ⁸	DL4: AIC100 at 7.5x 10 ⁸	DLTs	0			-	ORR at day 42 after infusion (%)	0	22		-	ORR for ATC pts (%)	0	50		-	DCR at day 42 after infusion (%)	0	56		-	DCR for PDTC pts (%)	0	60		-
	DL1: AIC100 at 1x 10 ⁷	DL2: AIC100 at 1x 10 ⁸	DL3: AIC100 at 5x 10 ⁸	DL4: AIC100 at 7.5x 10 ⁸																															
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Grade 3 Pneumonitis	0			2																															
Grade 1/2 CRS	0			10																															
Based on safety and efficacy, DL3: AIC100 at 5x 10⁸ was declared as RP2D; Cohort expansion at RP2D for ATC and PDTC will be initiated in 2Q'2025																																			
Implications																																			
- AIC100 shows promising efficacy and safety in treating aggressive thyroid cancers, with affinity tuning enhancing tumor targeting while minimizing side effects - Its built-in DOTATATE PET imaging enables real-time treatment monitoring and early positive results support further development and potential expansion to other ICAM-1-expressing tumors																																			

AUMOLERTINIB: P-III STUDY RESULTS



BRIA-IMT: LATE-BREAKER P-III STUDY RESULTS



Prespecified subset analyses of PFS based on biomarker status

- **DTH+** showed better PFS (4.47 vs 2.53 mos, $p = 0.001$)
- **Neutrophil-to-lymphocyte ratio (NLR) < 0.7 and ≥ 2.3** following the 1st treatment showed 3.07 mPFS ($p = 0.02$)
- **Circulating tumor cells (CTCs) < 1** showed better PFS (3.8 vs 2.4 mos, $p = 0.04$)
- **Baseline CAML count ≥ 5** showed higher mPFS (3.7 vs 2.2 mos, $p = 0.10$)



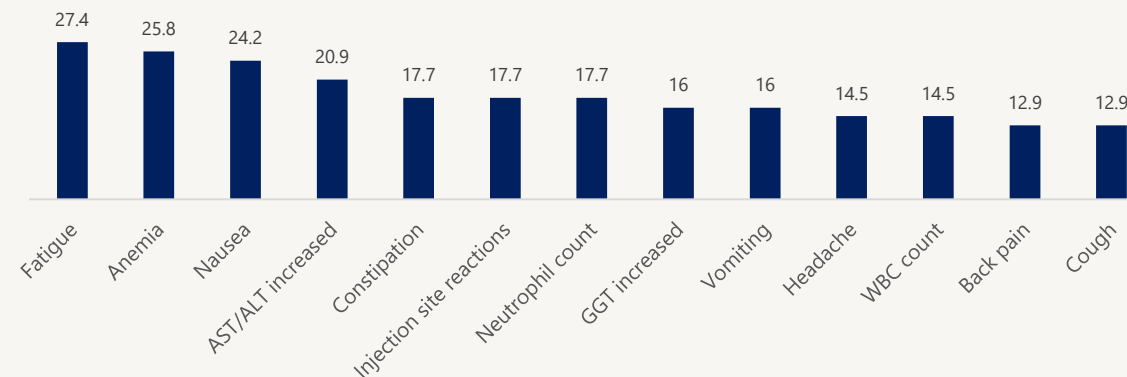
Bria-IMT, a combination immunotherapy comprising the allogeneic whole-cell vaccine SV-BR-1-GM

BRIA-ABC

Bria-IMT regimen (Day -2 or -3 Cyclophosphamide 300mg/m² Day 0 SV-BR-1-GM given intradermally divided into 4 inoculations Day 1-3 CPI infusion plus interferon administered intra-dermally within each SV-BR-1-GM inoculation site) + Retifanlimab vs TPC (N= 62)

	Bria-IMT With or Without CPI	Treatment of Patients'/Physicians' Choice (TPC)
Any HLA Match Median PFS (mos)	3.10	3.16
No HLA Match Median PFS (mos)	4.00	2.80
Hazard Ratio (HR)	0.77	2.226
p Value	0.59	0.6803
mPFS (mos)	3.67	

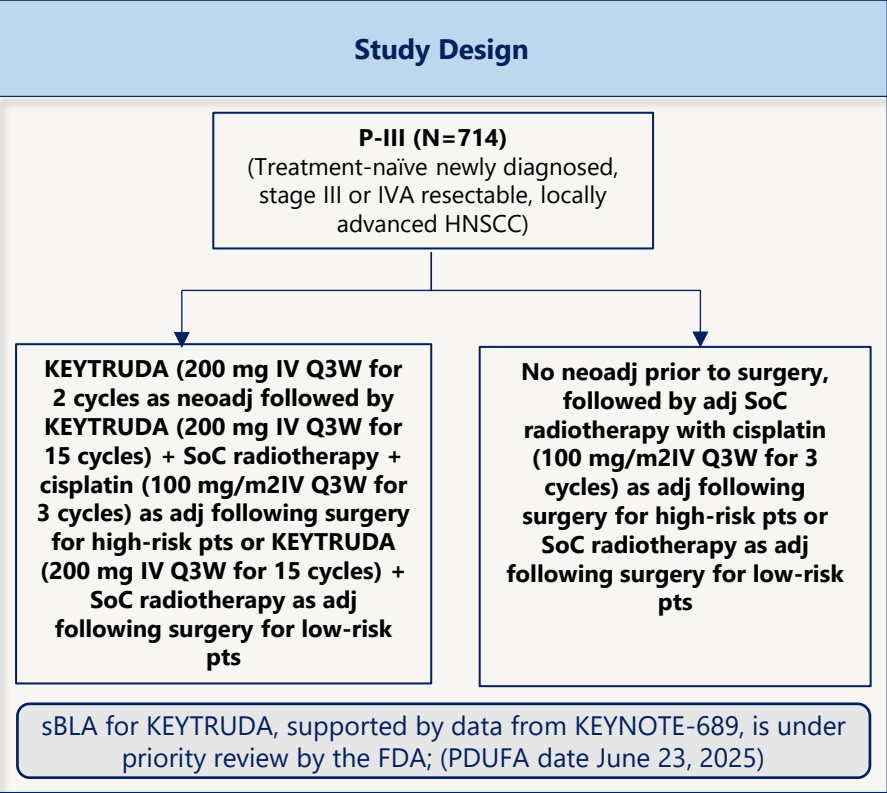
**Adverse events occurring in
> 10% of patients (%)**



Implications

- Positive P-III data and new biomarkers suggest their breast cancer therapy could see faster approval, attract investors, and gain a competitive edge in the market due to its safety and personalized approach

KEYTRUDA: P-III STUDY RESULTS



KEY INSIGHTS FROM AACR 2025

Antibody Drug Conjugate (ADC)	Next-gen ADCs with novel payloads and targets are broadening indications and improving efficacy in GI, breast, and lung cancers. Example: Mabwell's 7MW4911 (CDH17 targeting) shows early promise.
Bispecific/Trispecific Antibodies	Multi-specific antibodies (e.g., PD-1 + IL-12) are overcoming resistance and enhancing immune engagement in solid tumors.
Cell & Gene Therapies	Advances in CAR-T, NK, and TIL therapies are expanding to solid tumors, with improved engineering for potency and safety.
Targeted Therapies: RAS & Beyond	KRAS inhibitors (e.g., zoldonrasib, RMC-9805) achieve >60% response in NSCLC, reflecting progress against undruggable targets.
Early-Phase Clinical Breakthroughs	First-in-human data on HER2xCD3 bispecific, novel small molecules, and more-12+ new agents entering trials.
AI in Oncology	AI models now match experts in risk prediction, trial design, and patient selection, accelerating discovery and precision care.
Liquid Biopsy & ctDNA	ctDNA guides adjuvant therapy, detects minimal residual disease, and enables earlier, noninvasive monitoring of recurrence.
Cancer Vaccines	Personalized vaccines (e.g., NeoAg-VAX) show immune activation in cold tumors like MSS mCRC and glioblastoma.
Patient-Centric Innovation	Focus on survivorship, quality of life, and equitable access-bridging research and real-world patient benefit.

REACH US FOR IN-DEPTH REPORT



Himanshu Sehgal

Managing Partner

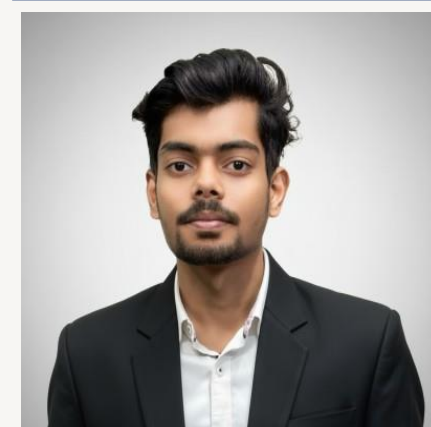


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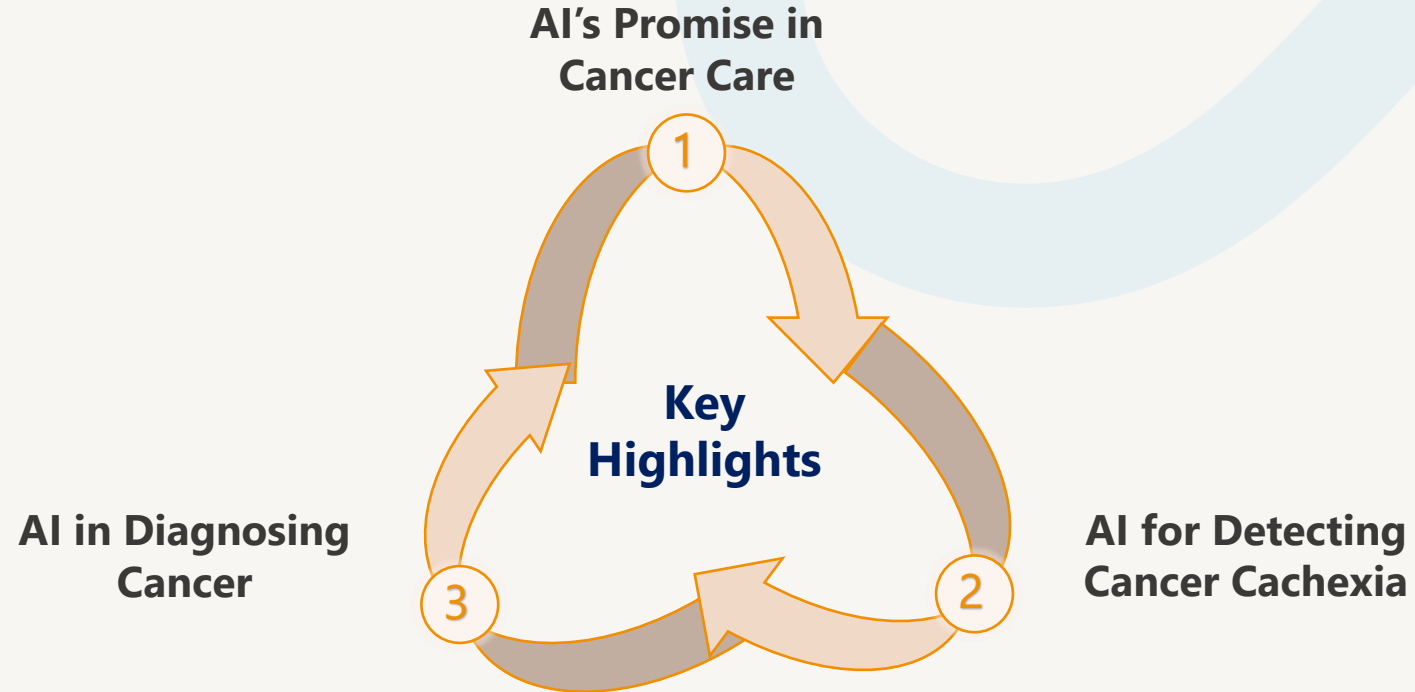
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IMPROVING CANCER CARE WITH ARTIFICIAL INTELLIGENCE



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